



Society of Petroleum Engineers

SPE-227701-MS

A New and Novel Managed Pressure Drilling System

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This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

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Abstract

Managed Pressure Drilling (MPD) is an adaptive drilling process used to precisely control the annular pressure profile throughout a well-bore. Drilling fluid is added to the annular by surface backpressure, which can be adjusted much faster in response to downhole conditions compared with conventional drilling method of changing the mud weight. Constant Bottom-Hole Pressure (CBHP) is a drilling methodology where the bottom hole pressure is kept at a constant value when the mud-pumps are on or off, i.e. when making a drill pipe connection. The constant bottom hole pressure method ensures well-bore stability is maintained during connection times.

The most crucial part of a Managed Pressure Drilling equipment package is the Rotating Annular pressure Control Device (RCD). The rotating annular pressure control device creates a constant annular seal around drill pipe allowing for annular pressure management by manipulation of the annular pressure from surface using flow chokes and a backpressure pump. The rotating annular pressure control device, however, also represents significant operating costs due to the requirement of regular changes and maintenance.

This paper describes a novel solution and alternative to a full managed pressure drilling equipment package that will trap mud returns in the drilling well-bore annulus to maintain a constant well bore pressure while making or breaking drill pipe connections as the mud pumps are shut down. As the technique is trapping pressure in the annulus opposed to using a closed loop pressurized mud returns system to precisely control the bottom hole pressure the overall system is a lot simpler, more robust and easier to maintain compared to a full managed pressure drilling system

The new solution can provide an estimated 70% of the well-bore pressure control possible with a full managed pressure drilling equipment package but at at 20% of the cost when compared to a full managed pressure drilling system. The reduced cost and reliability of the new system make an attractive alternative to using a full managed pressure drilling system for Constant Bottom-Hole Pressure applications.

Introduction – Managed Pressure Drilling

Managed Pressure Drilling (MPD) as defined by the International Association of Drilling Contractors (IADC) is an adaptive drilling process used to precisely control the annular pressure profile throughout the well bore. Managed pressure drilling utilizes a closed loop circulating system allowing drilling fluid to be added to the annular by surface backpressure. The addition of drilling fluid to the annulus to manipulate

the well bore pressure can be adjusted much faster in response to well bore downhole conditions compared with conventional drilling open loop circulation which requires changing the drilling fluid weight to control bottom hole pressure. The purpose of MPD is to manage the well bore annular pressure but also to understand the downhole pressure environment limits.

Constant Bottom-Hole Pressure (CBHP) drilling is a method where the well bore bottom hole pressure is kept at a constant value, at a specific depth, regardless of whether the circulation mud pumps are on or off. i.e. the bottom hole pressure is maintained constant between the dynamic circulating pressure (during drilling, reaming etc.) and the lower static pressure (when making or breaking connections pulling out of hole etc.). The constant bottom hole pressure method ensures well bore stability is maintained during connection times. (Zein 2017)

According to the International Association of Drilling Contractors the typical methods used to conduct constant bottom hole pressure drilling are: By keeping continuous circulation during drill pipe connections, by trapping annular pressure prior to shutting down circulation and by diverting mud pump flow rates across the well head. (IADC85). The use of an MPD equipment package is the most commonly used method of the three. Constant bottom hole pressure drilling is achieved by using the MPD equipment package to apply a certain amount of surface back pressure to the annulus to compensate for the difference in the circulating and static bottom hole pressure that occurs when circulation is stopped to enable a drill pipe connection to be made.

Managed Pressure Drilling Equipment

The three main equipment elements of a closed loop MPD equipment package that are required to facilitate the control of the annulus pressure are: A Rotating Annular pressure Control Device (RCD) a Pressure Control System and a Non Return Valve (NRV), as shown schematically in Figure 1. The function of the RCD is to create a seal between the drill pipe and the top casing inner diameter, which in a standard drilling operation is open to atmosphere (hence open-loop). The function of the NRV is to prevent any back flow from the annulus up the drill pipe. The pressure control system normally consists of an automated Choke Manifold, a flow measurement device (typically a Coriolis meter) and a Back Pressure Pump (some systems will not have back-pressure pump but use the mud-pump flow diversion to provide back pressure).

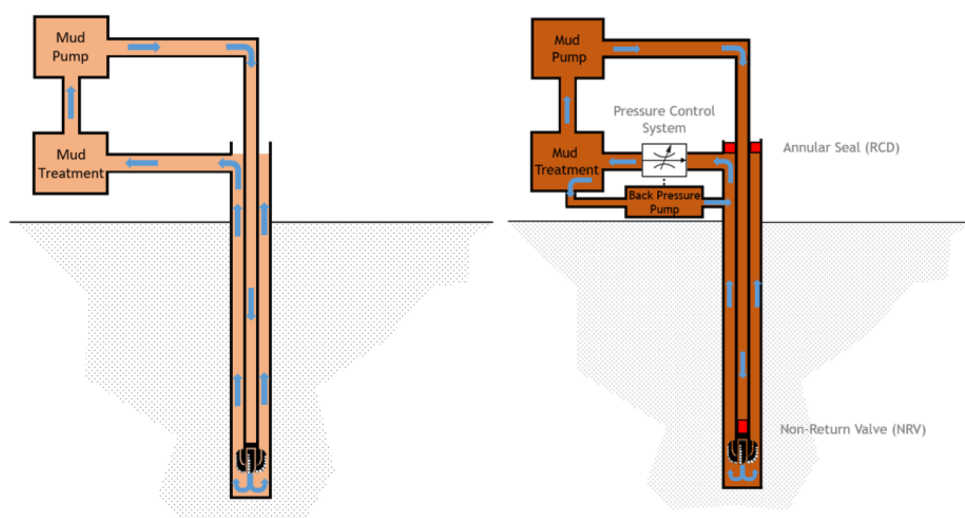


Figure 1—Simplified schematic of standard mud circulation (open-loop) and the addition of MPD (closed-loop).

Figure 1 shows a simplistic schematic of a conventional open loop drilling mud circulation system and an MPD closed loop system. It can be seen that the pressure in the annulus can be manipulated as required

with the pressure control system and back pressure pump by a combination of restricting the return flow from the annulus or by ‘topping- up’ the annulus using the back pressure pump when the main circulation mud pumps are off.

The Rotating Annular Control Device

The Rotating Annular pressure Control Device (RCD) is the key component in the MPD equipment package. The function of the RCD, as previously stated, is to seal the wellbore annulus, as shown schematically in Figure 1. As the RCD is containing the annulus pressure it is a primary well barrier and so considered a critical safety component, the manufacture, and testing qualifications are therefore covered by an American Petroleum Institute (API) specification, API 16RCD. According to API 16RCD the definition of an RCD is a "Drill-through equipment designed to allow the rotation of the drill string and containment of pressure using seals or packing elements that seal against the drill string (drill pipe, casing or other drill string components)" (Sehsah, 2020). The RCD is therefore required to seal around drill pipe during drilling operations and all drill-string movements, i.e. when the drill pipe is rotating, not rotating, moving downhole or moving out of hole. The RCD must therefore seal across a changing diameter and changing surface conditions as drill pipe tool joints are moved up or down through the RCD.

To accomplish the above functionality requirement an RCD has three main components: a seal element assembly, a bearing assembly and a housing. The bearing and seal element assemblies are typically retained inside the housing using a quick releasable locking mechanism (either locking dogs or a clamp). A simplified schematic of a typical RCD assembly is shown in Figure 2. The use of a quick release locking mechanism allows rapid removal and replacement of the bearing and seal element assemblies.

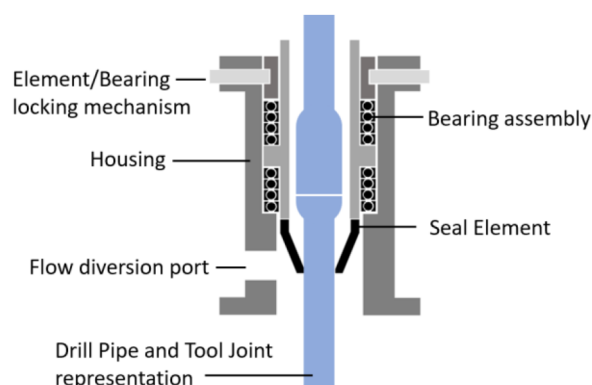


Figure 2—Simplified schematic of a typical RCD passive seal configuration.

The most common and simplest type of RCD seal element is a passive static friction type seal i.e. the seal element is a stretch fit seal with no active movement. As the passive element seal operates using friction it must rotate with the drill pipe, otherwise the pressure retaining seal would break and the seal surface would rapidly wear. For this reason, the seal element is attached to the bearing assembly which itself is anchored into the RCD housing. The bearing assembly allows the seal element to rotate with the rotation of the drill string. A typical passive seal element design is shown in Figure 3, the seal is normally constructed of a single elastomer with integrated wear and anti-extrusion features.

The two most common failures that occur in the RCD during operation are: The RCD static friction seal element leaks due to seal damage, or the RCD bearings fail. The failure of the bearings will rapidly also lead to a rapid seal failure (Machock, 2021).

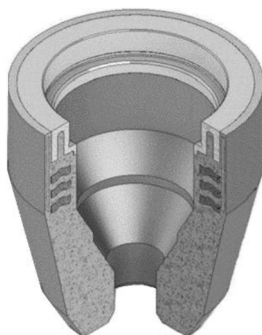


Figure 3—Sectioned Schematic of a typical RCD sealing element.

If we consider the operating requirements of an RCD element it is not surprising that the RCD static friction seal element fails, just considering the requirement of repeatedly passing drill string tool joints through the stretch fit friction seal is a tough operating ask which is made tougher by the often poor external surface condition of tool joints due to make up tong marks as shown in Figure 4. In addition, drill string tool joints can also have wear-resistant hard banding which adds an additional rough surface passing through the RCD seal element. Other drilling operating factors such as drill-string vibration, aggressive well fluids, drill string to RCD misalignment and operating temperatures are all reported factors that can significantly reduce the operating life of an RCD static friction seal element. (Sehsah, 2020).



Figure 4—Image of drill pipe tool joint tong marks.

A typical RCD static friction seal element failure is shown in Figure 5. The failure mode of the seal highlights the issue of when the RCD seal element fails it is usually a complete failure that occurs during drill string movement that then stops operations.

There are many RCD designs available today which have been optimized with respect to the design of seal element geometry, elastomer selection, and use of multiple seal elements. The RCD assembly configurations have been optimized to allow rapid removal and replacement of the RCD seal elements and bearings such that in the event of a RCD failure the downtime is minimal. RCD operations are also being subject to condition based monitoring (CBM) technologies such that seal element replacements can be predicted so that RCD operating conditions changed to be more efficient and seal element replacement can be scheduled before failure occurs and during downtimes (Machock, 2021). However, even with the optimized RCD solutions available today a large proportion of the MPD equipment package capital and operating costs can still be associated with the failure of the RCD element.



Figure 5—Image of typical RCD element failure (chunked).

Alternative designs for RCD's that use different sealing methodologies (i.e. not a passive static friction seal) are available, however, these solutions tend to be more complicated assemblies requiring more rig floor space and tend to be more expensive solutions. These alternative solutions are used in environments where regular maintenance and replacement of RCD seals are not possible.

A new alternative

An alternative method to using an RCD with a static friction seal that is continuously in contact with the drill pipe external diameter as it is rotated and run in and out of hole to achieve continuous bottom hole pressure drilling, is to use a dynamic seal element configuration that can be controlled to seal around the drill pipe external diameter as and when required. When an annular pressure seal on the drill pipe external diameter is not required the dynamic seal element retracts to a non-contact position. This type of seal design methodology can be applied to constant bottom hole pressure trapped pressure drilling methods to reduce or eliminate the change in bottom hole pressure during drill string connections i.e. when drilling has stopped, the drill string rotation has stopped, and circulation has stopped.

The dynamic seal element can be activated once movement (rotation and displacement) of the drill string has stopped just before bleeding down the circulation is complete in order to trap the higher annular pressure created due to circulation. Once a connection has been made and circulation starts to ramp up the seal element can be released away from the drill pipe and tool joint external diameters and drilling can continue. The functionality requirements for such trapped pressure methodology are therefore a fast-acting dynamic seal element system that has a relaxed diameter sufficiently large to pass drill string tool joint connections through without contact but with an activated seal diameter suited to the drill pipe body external diameter. The seal system developed to fulfil these requirements is called the Trapped Annulus Pressure Device (TAP-D).

The TAP-D system is configured in a similar way to a standard RCD system and consequently looks very similar, but in place of a RCD seal element and combined bearing assembly locked into the RCD housing there is a dynamic seal element assembly locked into a similar housing as shown schematically in [Figure 6](#).

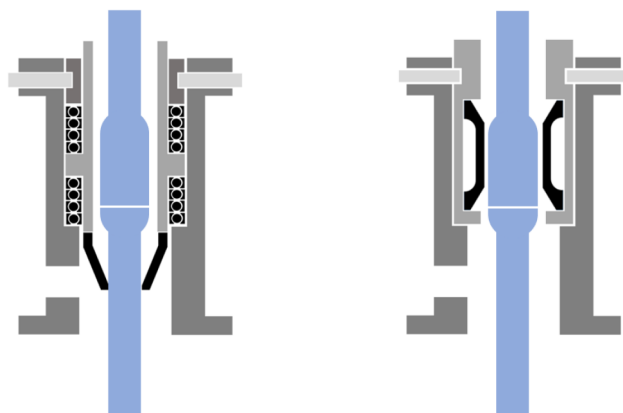


Figure 6—Simplified schematic of a typical RCD configuration and the TAP-D configuration.

The TAP-D system is to be installed on the wellhead in the same manner as a standard RCD assembly. Figure 7 show the two operating states of the TAP-D: A seal element relaxed state when drill pipe and tool joints pass through with no contact and a seal element activated state when the drill pipe is no longer rotating and annular flow is diverted to the TAP-D choke control.

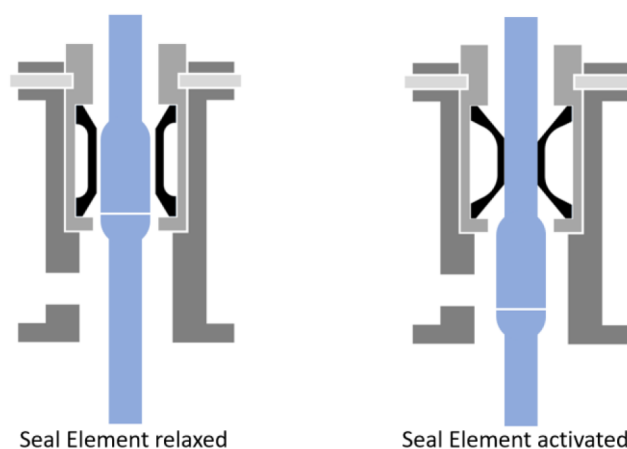


Figure 7—Simplified schematic of the TAP-D relaxed and activated

Figure 8 shows the actual TAP-D assembly and seal element. The TAP-D seal element assembly design intent was to keep the assembly as simple as possible allowing for ease of assembly, disassembly, a minimum number of parts and therefore ease of maintenance and low cost. The seal element assembly consists of just three subassemblies held within a housing: each sub assembly is slid into the housing and a housing cap then fitted. This simple yet functional design allows for a minimal spare requirement at the rig site and rig site redress of a seal element assembly is possible if required.

The dynamic TAP-D seal element is hydraulically activated and controlled through a simple valve assembly and hydraulic feed, the same hydraulic feed and control is used to activate the seal element assembly quick release locking dog system.

The TAP-D is coupled with a semi-automated controller and choke manifold and the complete equipment package is referred to as a 'Compact MPD' system. The semi-automated controller will precisely monitor and control the back pressure as the mud pumps are manually ramped down/up before and after making pipe connections.

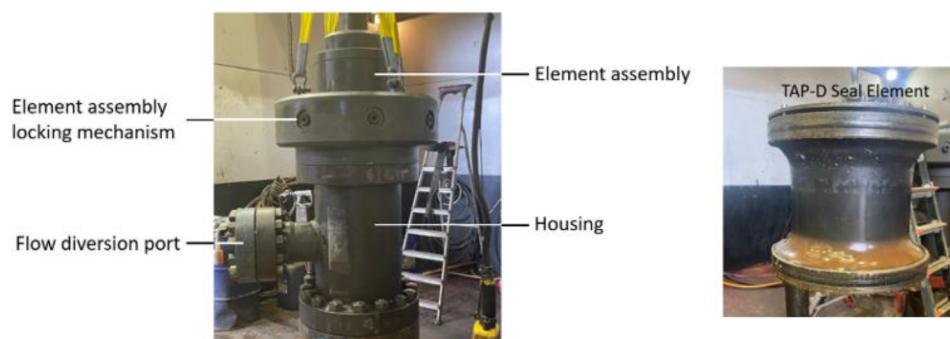


Figure 8—TAP-D complete assembly and TAP-D Seal Element

As the Compact MPD operating technique is to trap well bore pressure in the annulus opposed to a full closed loop MPD system utilizing pressurized mud returns to control the bottom hole pressure the overall Compact MPD system is simpler, requires less operating equipment, is more robust and easier to maintain.

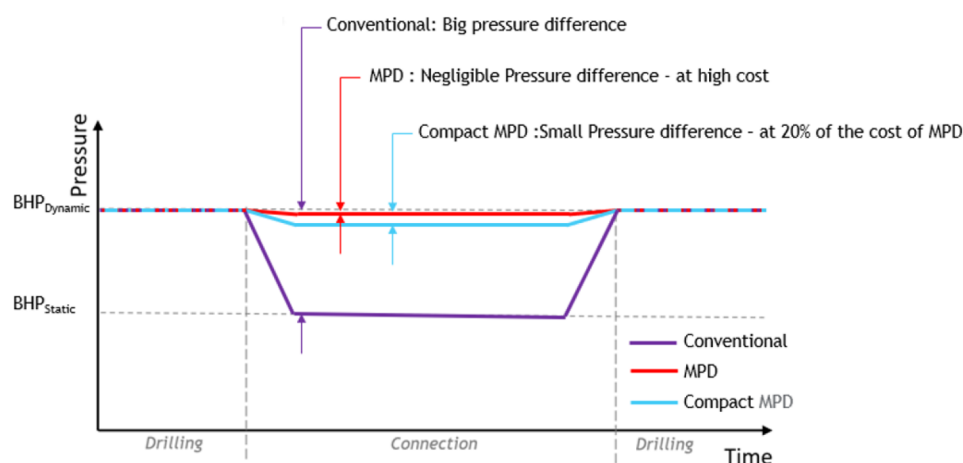


Figure 9—Schematic representation of the Static and Dynamic bottom hole pressure changes for Conventional, MPD and Compact MPD drilling operations during a Drill pipe connection.

Conclusions

A typical MPD equipment package consists of an RCD, a choke manifold, an auxiliary back pressure pump with a pump manifold, and Coriolis flow meters. The Compact MPD equipment package replaces the standard equipment package with a single TAP-D installed on the wellhead in the same way as a standard RCD with just a simple choke manifold. The Compact MPD equipment package is cheaper, requires less installation and rig-up/rig-down times. The system requires less ‘hands on’ maintenance and due to the dynamic seal element design eliminates the requirement for a bearing assembly and does require frequent replacement of the rubber seal element during a drilling operation. As a result, it is estimated the cost of operating the Compact MPD system will be +/- 20% of the cost of operating a standard MPD equipment package and the system can achieve +/- 70% of the drill pipe connection time bottom hole pressure stability achievable utilizing a standard MPD equipment package. A comparison of the drilling methodologies considering changes in bottom hole pressure during making a drill pipe connection is shown schematically in Figure 9. In conventional open loop circulation drilling during a drill pipe connection the bottom hole pressure will drop from the dynamic circulation pressure to the static pressure as the mud circulation is stopped and will remain at the static pressure until circulation is initiated again once the drill pipe connection has been made. A standard MPD equipment package by utilizing flow chokes and backpressure application to the annulus can compensate for the change in pressure as circulation is stopped

with a negligible difference between the drilling and non-drilling bottom hole pressure. The Compact MPD system by trapping the annulus pressure during circulation bleed down can compensate an estimated 70% of the dynamic to static pressure change. (Compact MPD cannot completely compensate for the dynamic to static pressure drop as it has no capability to top up the annulus pressure.)

The Compact MPD package has been designed for constant bottom hole pressure applications where the precise and full functionality of a standard MPD equipment package is not required but where the difference in static and dynamic bottom hole pressure with conventional open loop circulation drilling will potentially create issues (such as losses, stuck pipe, bore hole stability etc.). The lower system cost, simple installation and lower maintenance requirements of the Compact MPD system make it a potential for using constant bottom hole pressure drilling techniques where the cost of a full MPD equipment package is prohibitive to its use.

Abbreviations

API	American Institute of Petroleum Engineers
BOP	Blow Out Preventer
CBHP	Constant Bottom Hole Pressure
CBM	Condition Based Monitoring
IADC	International Association of Drilling Contractors
NRV	Non Return Valve
MPD	Managed Pressure Drilling
RCD	Rotating annular pressure Control Device
TAP-D	Trapped Annular Pressure Device

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